

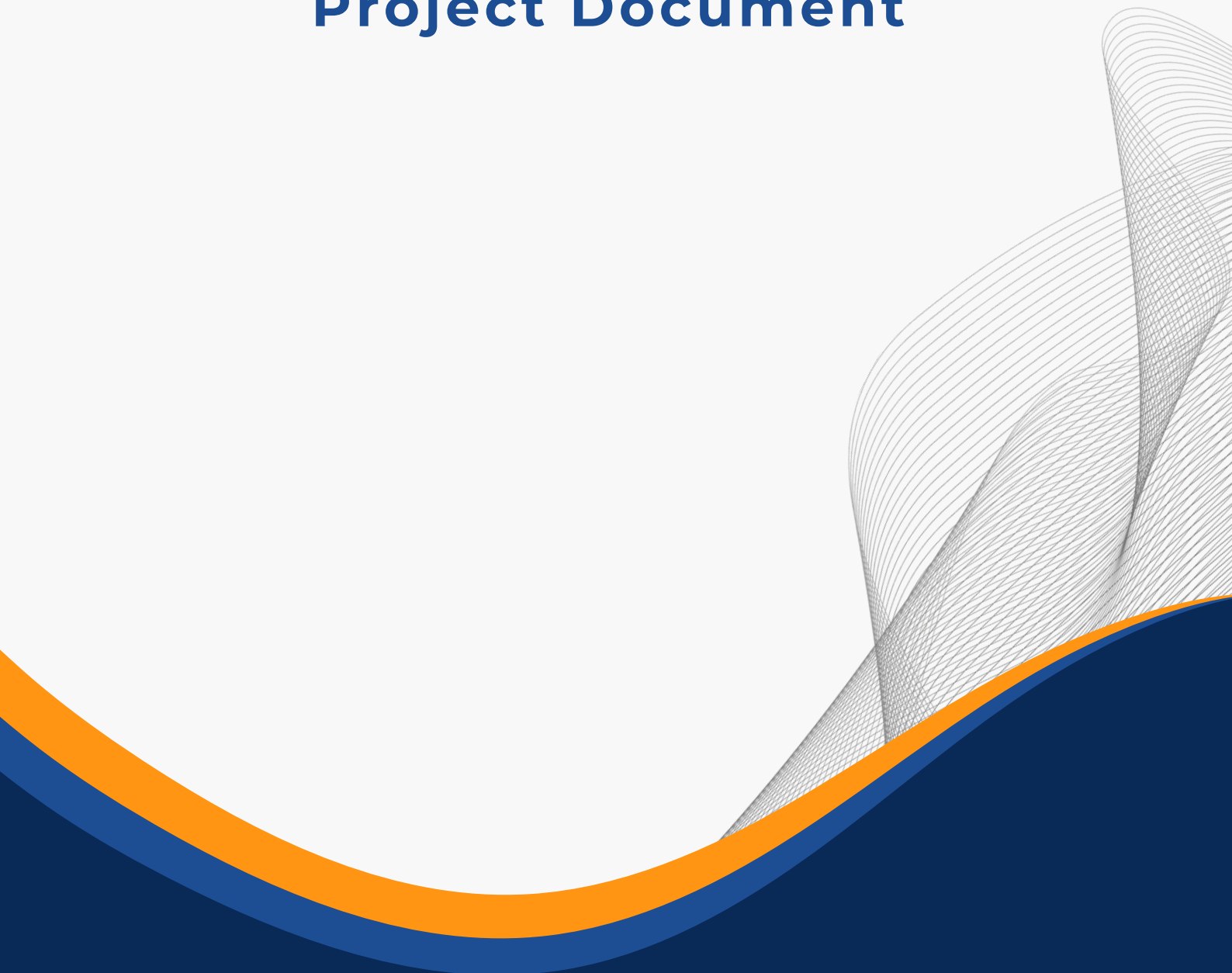


CTEC22043

Faculty Management System

Group H

Project Document



1. Executive Summary

The Faculty Management System (FMS) is a desktop-based Student Information System (SIS) designed for the Faculty of Computing & Technology. Developed in Java using the Java Swing framework, the system aims to streamline academic administration by providing distinct portals for three core roles: Administrators, Students and Lecturers.

FMS eliminates manual academic scheduling and paperwork by centralizing profile management, weekly timetables, course enrollment registry, and digital lecture material distribution.

2. System Scope & Core Features

The application fulfills the following 9 administrative and user requirements:

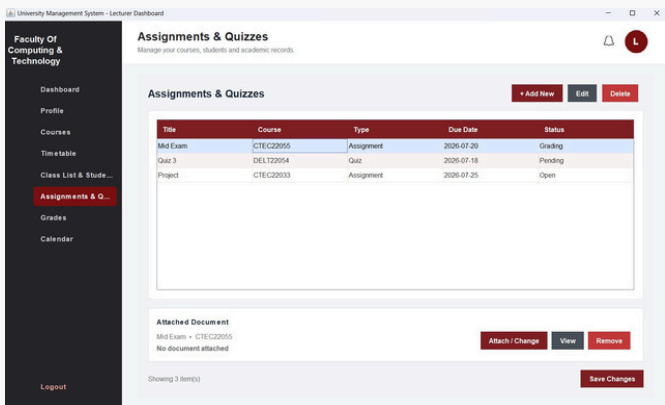
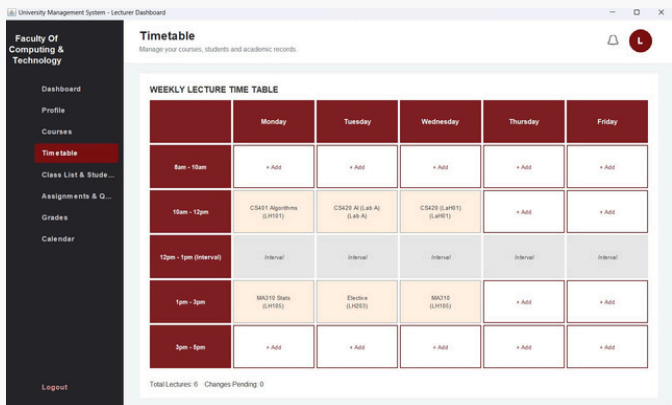
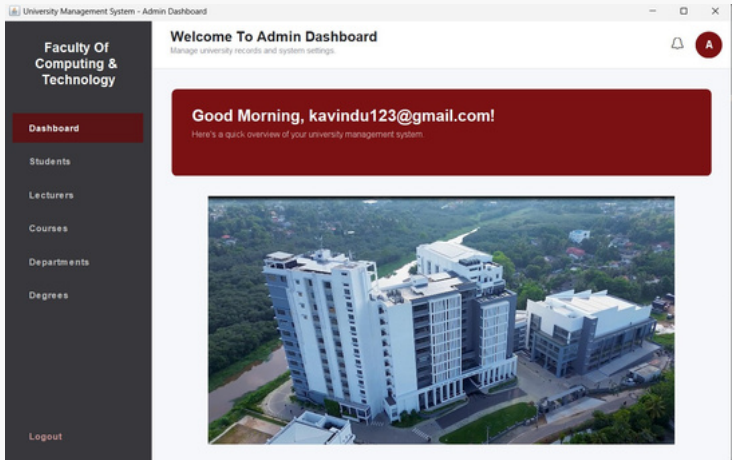
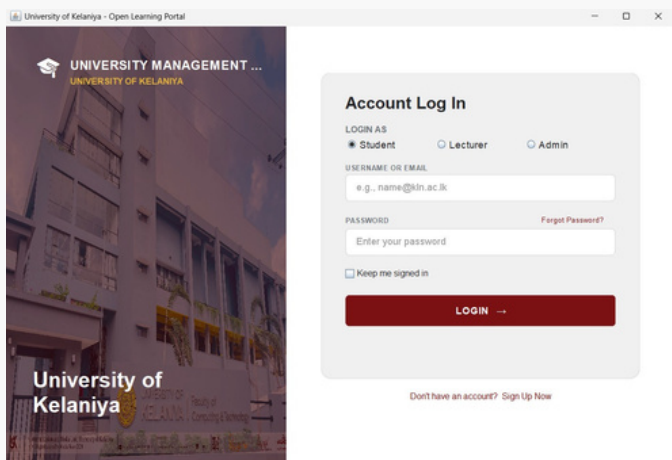
1. Authentication (Sign In / Sign Up): Secure login screens with physical role selectors (Admin, Student, Lecturer).
2. Profile Details: Editable records storing full names, emails, IDs, and phone numbers.
3. Weekly Timetable Grid: A visual weekly dashboard showing class hours (08:00 - 17:00) mapped from Monday to Friday.
4. Course Enrollments: Tracks student course pathways and registry lists.
5. Student Portal Workspace: Interface for students to update profiles, view personalized schedules, and check enrolled courses.
6. Lecturer Portal Workspace: Interface for academic staff to manage teaching schedules.
7. Lecture Materials Manager: Allows lecturers to upload file pathways (PDF, Video, Slides) and link them directly to the course syllabus.
8. Department Registry: Administrative module to view and organize academic departments.
9. Degree Registry: Administrative module to configure and manage degree programs.

3. Technical Architecture

- Frontend: Java Swing (using modern custom paint rendering for gradients,roundedborders and flat-designed table structures).
- Development IDE: IntelliJ IDEA / Eclipse.
- Architecture Pattern: Model-View-Controller (MVC) to separate visual screens (Views) from the SQLite database connections (Models).

*GitHub Organization:
OOP-Assignment-Group-H
University-Management-System

*Repository:
<https://github.com/OOP-Assignment-Group-H/University-Management-System>



4. Team Contribution Registry

Below is the formal registry of responsibilities and code contributions for the key developers of this project:

Developer Contributions Table

Student Registration No.	Core Assigned Modules	Technical Responsibilities
CT/2022/073	Project Initializer, Sign Up, Admin Core Entities	Setup Initial Project Structure, Main Execution Entry Point, Sign Up Visual Form, Admin Course Management Panel, Admin Degree Planner, Admin Department Registry
CT/2022/062	Sign In Module, Admin Dashboard, Scheduling	Sign In Form Interface, Admin Main Dashboard Layout Frame, Visual Weekly Timetable Grid Module
CT/2022/077	Student Workspace	Student profile form, Save action logic, Student-side course view
CT/2022/033 CT/2022/052	Lecturer Workspace	Lecturer profile, Syllabus manager, Lecturer-side class list
CT/2022/077	Material Uploader	FileChooser setup, Database directory logging, material tables
CT/2022/014	Database Integration	Local SQL connection pool, SQLite helper class, DB schema integration